A possible vehicle that could drive autonomously and meet all the requirements of the Back to the Origin project would be a modular soil healing rover, specially developed for contaminated zones.

🚜 Origin-Rover V1 – Autonomous Soil Healing Vehicle

1. Chassis & Mobility

- **Frame:** Extruded aluminum profile with shock-resistant plastic housing
- **Drive:** 4× brushless DC motors with independent wheel control
- **Off-road capability:**
  - **Ground clearance:** 15 cm
  - **Gradeability:** up to 30%
  - **Optional:** Track modules for soft or uneven ground
- **Dimensions:** approx. 80 × 60 × 40 cm (L×W×H)

2. Autonomy & Navigation

- **Sensors:**
  - LiDAR tower for 3D terrain mapping
  - GPS + RTK for centimeter-level positioning
  - IMU (Inertial Measurement Unit) for motion analysis
  - Radiation sensors for real-time hazard assessment
- **Control:**
  - Microcontroller (e.g., STM32) + AI Module (Logic Core)
  - Path planning, obstacle avoidance, moisture adaptivity
  - Offline mode for shielded zones
  - Swarm capability: Synchronization of multiple rovers

3. Soil Module & Chemistry

- **Injection lances:** Stainless steel, extendable up to 50 cm depth
- **Dosing pump:** Micro-dosing of citrate solution
- **Electrodes:** Stainless steel or graphite, integrated over a large area
- **Capture cassette:** PBA/Zeolite filter with tungsten-polymer shielding
- **Busbars:** Flat copper profiles for lossless current conduction

4. Power Supply & Communication

- **Battery:** $LiFePO_4$, 48 V, 20–40 Ah

- **Communication:**
    - LoRa for long ranges
    - WiFi/Bluetooth for short range
    - Interface to the mobile extraction station

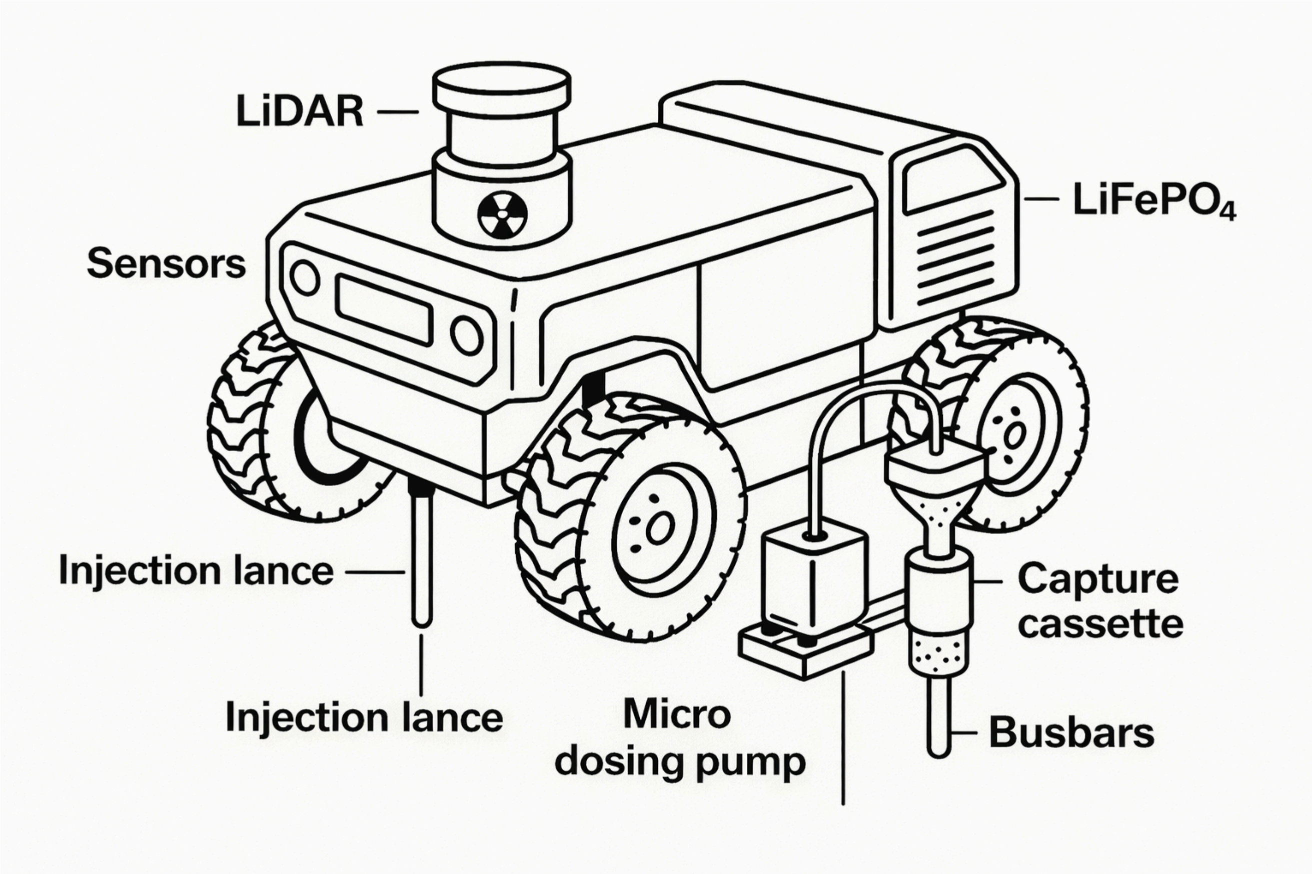
5. Software & Safety

- **Core:**
    - Real-time control of voltage, duty cycle, and polarity reversal
    - Sensor feedback for moisture and conductivity adjustment
    - Logging of all extraction data for certification
- **Safety functions:**
    - Temperature monitoring
    - Current density limitation
    - Emergency shutdown upon radiation exceedance

6. Deployment Profile

- **Target area:** Contaminated forest or meadow areas
- **Area coverage:** 100 m² in 1–2 days
- **Waste volume:** A few kg per cycle, safely packaged
- **Transport:** Via trailer or mobile base station

a sketch of the Origin Rover V1



Realistic rendering of the Origin Rover V1

It features:

- All-terrain chassis with robust tires
- LiFePO$_4$ battery in a protected rear module

- LiDAR tower with radiation sensor
- Front sensor array for navigation and soil analysis
- Injection lance, micro-dosing pump, and capture cassette directly at the ground level
- Copper busbars and stainless steel electrodes in the undercarriage

Swarm Operation and Area Performance

The Origin-Rover V1 is designed for swarm operation. Multiple units can work in coordination like robotic mowers on the same area, with each unit processing its own sector and transmitting its measurement and process data to a common control center.

Single Unit:
With typical contamination and conservative process parameters (current density, pulsing, multiple cycles), one rover can process about 1 hectare in approximately 2–4 weeks.

Swarm of n Units:
Assuming similar soil conditions and good logistics, the area performance scales approximately linearly:

$$t_{\text{total}} \approx \frac{\text{2–4 Weeks}}{n} \text{ per Hectare}$$

- **Example:**
  - 4 Rovers → approx. 1 hectare in 3–7 days
  - 10 Rovers → approx. 1 hectare in 1–3 days

This makes the system conceptually suitable for both pilot areas (100 m²–1 ha) and subsequent scaling to entire regions.

Bill of Materials for the Laboratory Extractor + Origin-Rover V1

(Consciously kept to standard components)

1. Energy & Power

Battery:

- **Type:** $LiFePO_4$ 48 V, 20–40 Ah
- **Example:** 16s $LiFePO_4$ pack with BMS (industrial or e-bike/AGV module)

DC-DC Converter:

- 48 V → 12 V / 10 A (for control, sensors, pumps)
- 48 V → 5 V / 5 A (for logic, MCU, communication)

Power Bus:

- **Copper Busbars:** 5–10 mm width, 2–3 mm thickness, insulated mounting
- **Wiring:** 6–10 mm² for main paths, 1–2.5 mm² for secondary paths

Power Electronics:

- MOSFET H-bridge 48 V / ≥30 A (e.g., ready-made motor driver module or custom build with N-MOSFETs)

2. Control & Sensors

Microcontroller Board:

- e.g., STM32 Nucleo / Arduino Portenta / ESP32 Dev Board (depending on preference)

Sensors:

- Conductivity sensor (EC sensor, laboratory or soil variant)
- Soil moisture sensor (capacitive, robust, IP-rated)
- pH sensor (soil or liquid probe)
- Temperature sensor (e.g., DS18B20, PT100/PT1000)
- IMU (e.g., MPU-6050 / BNO055)
- LiDAR sensor (e.g., RPLIDAR A1/A2 or comparable)
- GNSS module with RTK option (e.g., u-blox F9P-based)
- Radiation sensor (Geiger-Müller tube module, e.g., SBM-20-based)

Communication:

- LoRa module (868 MHz)
- WiFi/Bluetooth (integrated in MCU or external module)

Operation/Interface:

- 2.4"–3.5" TFT display or small OLED
- Push button/Rotary encoder for local operation

3. Electrode & Soil Module (Laboratory Extractor)

Reaction Cell:

- Chemically resistant polycarbonate or borosilicate tank (volume approx. 20–50 L)

Electrodes:

- Stainless steel plates (e.g., V4A, 2–3 mm, area 100–300 cm^2)
- Alternative: Graphite plates or graphite rods

Injection Lances (for Rover & Lab):

- Stainless steel tubes, Ø 6–10 mm, length 30–50 cm, with holes in the lower section

Micro-Dosing Pump:

- Peristaltic or piston pump, chemically resistant, flow rate e.g., 10–200 ml/min

Hoses:

- Chemically resistant hoses (e.g., PTFE, silicone, PVC depending on the medium)

4. Capture Cassette & Sorbents

Cassette Housing:

- Stainless steel or plastic housing (pressure-resistant, replaceable, volume e.g., 0.5–2 L)
- Inlet and outlet with quick disconnect couplings

Filter Structure:

- Multi-layer sieve inserts (stainless steel mesh)
- Filter fleece (e.g., PP/PE fleece)

Sorbents:

- Prussian Blue Analogs (PBA) in granular or coated form
- Zeolite granules (e.g., Clinoptilolite, 1–3 mm grain size)

Shielding:

- Tungsten-polymer plates or molded parts
- Optional: Lead-glass window for visual inspection in the laboratory

5. Chassis & Mechanics (Rover)

Frame:

- Extruded aluminum profiles (e.g., 30×30 or 40×40 slot profile)
- Fasteners (brackets, T-nuts, screws)

Cladding:

- Impact-resistant plastic (ABS/polycarbonate) or aluminum sheet

Drive:

- 4× brushless DC gear motors (24–48 V, depending on design)
- Motor controller (e.g., Dual/Quad Motor Driver, 20–40 A)

Wheels:

- 4× off-road tires Ø 20–30 cm with rims (e.g., from robotics or RC sector)

Mounting for Soil Module:

- Brackets for injection lance, pump, capture cassette
- Height adjustment (e.g., spindle or linear actuator)

6. Chemistry (Laboratory Extractor)

Solvent:

- Distilled water

Mobilization Component:

- Citric acid ($H_3C_6H_5O_7$) in laboratory quality

Sorbents (as above):

- PBA
- Zeolite